Bootstrap Framework

Designing Responsive Web Applications

- Open Source
- Free

Build once • Adapt everywhere • Better user experience



What is Bootstrap & Problem

1. Introduction

A popular, open-source framework for building responsive, mobile-first websites with less code.

Easy to use

2. The Challenge

How can we create websites that work properly across multiple devices without messy custom CSS?

3. The Solution

Bootstrap provides a responsive grid system, styling classes, and reusable UI components that adapt to any screen size.

✓ Faster Development + Consistent UI + Better Experience

4. Key Features & Grid

Responsive Grid System
Uses rows & columns for breakpoints

Mobile-First Design
Scales up from small screens

Prebuilt Components
Ready-made nav, buttons, cards

Utility Classes
Quick spacing, colors, & typography

Faster Development
Highly reusable styles

5. How It Works

HTML Structure	→	Bootstrap CSS & Grid	→	JS Plugins	→	Responsive Web Application
Content and page elements		Styles, columns and breakpoints		Interactive components		Adapts to desktop, tablet and mobile

6. Responsive Grid Example

```
<div class="container">
  <div class="row">
    <div class="col-12 col-md-6 col-lg-4">
      Column 1
    </div>
    <div class="col-12 col-md-6 col-lg-4">
      Column 2
    </div>
    <div class="col-12 col-md-6 col-lg-4">
      Column 3
    </div>
  </div>
</div>
```

How it works?

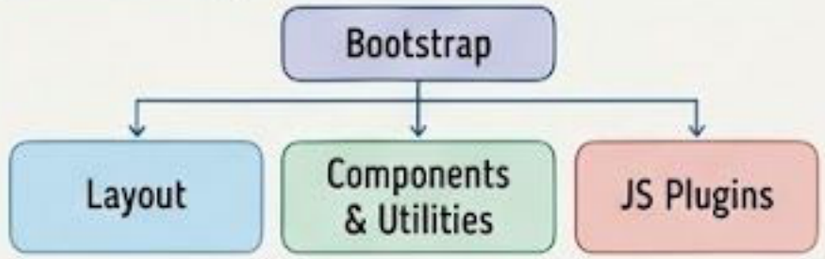
Screen size	Columns per row
Mobile	1 col
Tablet	2 cols
Desktop	3 cols

Bootstrap Grid Classes:

- col-12 = 100% width
- col-md-6 = 50% width (medium+)
- col-lg-4 = 33.32% width (large+)

Architecture & Responsive Design

7. Bootstrap Architecture



8. Design in Action



9. Advantages & Limitations

✓ Faster development
✓ Consistent design
✓ Reusable components

- Default designs can look similar
- Customisation may require extra CSS

10. Real-World Applications

- Portals & websites
- Admin dashboards
- E-commerce interfaces

11. Conclusion

Bootstrap simplifies responsive web development with a mobile-first grid and prebuilt styles.

One codebase, multiple screens!

References

1. Official Bootstrap Documentation – https://getbootstrap.com/#ecs/
2. Bootstrap Grid System – https://getbootstrap.com/docs/5.3/layout/grid/
3. Bootstrap Components – https://getbootstrap.com/docs/5.3/components/

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